

R Project Setup and Working Directory Guide

Before we begin any coding, we need to make sure your files are organized correctly. Good file organization will save you a lot of confusion later.

1. Set up your course folder

For this course, you should keep all course materials in one main folder on your computer.

1. Create a main course folder

Create a new folder and name it:

`Advanced_data_analysis_course`

2. Check file names carefully

File addresses (paths) and names in R must match exactly.

- uppercase and lowercase letters matter
- underscores and spaces matter
- punctuation matters
- file extensions matter

For example:

(notice the underscores and the capitalization of only the first word!)

`Advanced_data_analysis_course`

is *NOT* the same as:

`advanced data analysis course` [space instead of underscore]

`Advanced Data Analysis Course` [spaces and capital letters]

`ADVANCED DATA ANALYSIS COURSE` [capital letters]

`Advanced_data analysis_course` [mix of space/underscore]

...and so on...

3. Create a subfolder for this workshop

Inside that main folder, create a second folder called:

`pre-course refresher`

Your folder structure should look something like this:

`Advanced_data_analysis_course/
└─ pre-course refresher/`

4. Move your downloaded files

You downloaded these two files from Google Drive into your Downloads folder:

`Radac pre-course refresher.Rmd`

`Palmer Penguins Raw.csv`

Move both files from Downloads into the pre-course refresher folder.

Your folder should now look like this:

`Advanced_data_analysis_course/
└─ pre-course refresher/
 └─ Radac pre-course refresher.Rmd
 └─ Palmer Penguins Raw.csv`

2. Create an R Project

We will use an R Project for this course. This is important because it helps RStudio automatically keep track of the correct working directory.

1. Open RStudio
2. Go to: File > New Project
3. Choose: Existing Directory
4. Navigate to your main course folder: `Advanced_data_analysis_course`
5. Click Create Project

RStudio should now reopen inside your course project.

3. Open the refresher file from inside the project

In the Files pane, click into the `pre-course refresher` folder
Open `Radac pre-course refresher.Rmd`

You should now be working inside the correct project.

4. What is a working directory?

Your working directory is the folder R is currently using to look for files and save output. Because you opened an R Project, your working directory should now be:

`.../Advanced_data_analysis_course`

That means your .csv file is inside a subfolder called:

`pre-course refresher`

5. Check your working directory

Run:

```
getwd()
```

Your output should end with something like:

```
.../Advanced_data_analysis_course
```

If your working directory is *NOT* the main course folder, set it manually like this:

```
setwd("~/Advanced_data_analysis_course")
```

Then check again:

```
getwd()
```

6. Check that R can see your files

Run:

```
list.files()
```

You should see:

```
pre-course refresher
```

Now run:

```
list.files("pre-course refresher")
```

You should see these files:

- `Radac pre-course refresher.Rmd`
- `penguins.csv`